Rotocaster - screws by part

v18. Everything that bolts to each printed part. T-nut = drop-in M5 nut in the rail slot.

FRAME gusset brackets, bought — 8 corners

M5 x 8	16	two per gusset, one into each rail
M5 T-nut	16	one per screw

01 POST bearing holder — x2

M5 x 35	4	two per tower, front-to-back clamp onto the post. Crosses 26.2 mm of plastic before the post slot.
M5 x 16	4	two per tower, side-to-side clamp, at 90 deg to the pair above. Insert from the INBOARD face (facing the frame) - on the right tower the upper head would otherwise overlap part 13's base by 0.25 mm.
M5 T-nut	8	one per clamp screw
608ZZ	2	one per tower, press fit into the outboard face

02 / 02b SHAFT gripper — x4

M5 x 25	8	two per gripper, down through 16 mm into the rail slot. Same for either version.
M5 T-nut	8	one per screw
M4 x 40	4	02b ONLY - the pinch screw closing the split bore. Head in the counterbore, 32 mm grip plus the nut. Not needed on the D-flat version.
M4 nut	4	02b only, one per pinch screw

03 OUTERFRAME top bridge, Motor 2 — x1

M5 x 16	2	one into each rail end, through 6.7 mm
M5 T-nut	2	one per screw
608ZZ	2	lower seats from underneath, upper from the top face of the plate
M4 x 25	4	carries part 14. Shared with part 14 below - do not double count.
M4 nut	4	underneath the plate, clear of the pillar

04 OUTERFRAME bottom bridge — x1

M5 x 16	2	one into each rail end
M5 T-nut	2	one per screw
608ZZ	1	from underneath

05 POST motor 1 spacer — x1

M5 x 45	4	into the tower, entering from its INBOARD face. Threads into the nuts below.
M5 nut	4	trapped in the hex pockets in this part's legs. Drop them in before you offer it up.
M3 x ?	4	Motor 1 onto the 6 mm top plate, into the motor's tapped holes. LENGTH = 6 + usable thread depth. Your M3x12 and x10 were too long - measure how far an M3 goes into the bare motor first.

Rotocaster - screws by part

Continued.

06 INNERFRAME plate shelf — x2

M5 x 16	4	two per bracket, horizontally into the inner rail. Set the height before tightening.
M5 T-nut	4	one per screw
M4 x 25	4	two per bracket, up through the slots into the plate. Shared with part 08 - do not double count.
M4 nut	4	underneath. Nip only - these slots are your balance adjustment.

07 OUTERFRAME balance weight — x2

M5 x 55	4	two per weight, down through 45 mm into the bottom rail slot. Position at x = +/-120 mm.
M5 T-nut	4	one per screw
M8 nut	79	NOT a fastener - this is the ballast, about 204 g per pocket. Loose nuts, poured in.

07b balance weight cover — x2

M3 x 12	8	four per cover, self-tapping into the printed bosses. Load the ballast first.

08 MOULD plate — x1 (whole, at 45 deg)

M4 x 25	-	see part 06 - same four screws
M6 x ?	4	mould foot onto the plate, 140 mm PCD, counterbored. LENGTH NOT SET - depends on your mould's flange, which does not exist yet. Plate gives 3.5 mm after the counterbore.
M6 washer	8	both ends of each screw. This joint carries the whole mould and gets undone every mould change.

10 MOULD lid — x4 quarters

M5 x ?	6	lid onto the mould rim, 280 mm PCD. LENGTH NOT SET - mould dependent. Tighten in a star pattern, not around the circle.
M5 washer	12	both ends of each
M4 x 20	3	pour plug onto the lid, 58 mm PCD. Through 8 mm of plug head plus 5 mm of lid. Fit the O-ring first.
M4 nut	3	one per plug screw

12 SLIPRING shaft-to-stub coupler — x1

M3 x 12	2	two pinch screws, one on the shaft end, one on the rotor stub. Self-tap into the far jaw - no nut. Light load: only the slip ring's own brush friction.
HOLLOW SHAFT	1	the RIGHT trunnion must be 8 mm OD tube, 3 mm bore, 120 long. The six rotor wires run down its middle to Motor 2. Thread them through part 12's centre hole and into the tube BEFORE fitting.
USE 02b	1	on that tube, not a ground D-flat - a 1 mm flat would cut the 2.5 mm wall to 1.5 mm.

13 SLIPRING flange bracket — x1

M5 x 45	2	onto the RIGHT tower's outboard face, the free 56 x 46 pattern. Two diagonal corners are enough. Base 6 mm plus tower 34 mm.
M5 nut	2	on the tower's inboard face, with an M5 washer
M3 x 12	3	flange onto the plate's OUTBOARD face, through the radial slots. Slip ring goes in from outside, stub first, through the 23 mm bore. Drive the screws from outside; rotate the ring until holes line up.
M3 nut	3	on the plate's inboard face - reach them through the open sides

14 MOTOR2 coupler standoff — x1

M3 x ?	4	Motor 2 onto the 4 mm cap. LENGTH = 4 + usable thread depth - likely M3x8. Into the motor's tapped holes - no nut. Drive them UP from inside, through the open underside, BEFORE this part goes onto part 03.
M4 x 25	4	the feet, down into part 03's plate. Same four listed under 03.

Running totals

M5: x8 = 16 x16 = 12 x25 = 8 x35 = 4 x45 = 6 x55 = 4
M4: x20 = 3 x25 = 8 x40 = 1 min (02b on the right tube; 4 if all grippers are 02b)
M3: x12 = 13 (covers, slip ring) + 8 motor screws, length set by thread depth
M6: x4, length still to be decided
Nuts: M5 T-nut 44 M5 hex 6 M4 hex 11 (+1 per 02b) M3 hex 3 M8 ballast 79
Washers: M5 x14 M6 x8 Bearings: 608ZZ x5
Also: 8 mm OD tube, 3 mm bore, 120 long, for the right trunnion